

Model Lock-In Sheet

ECO 317 – Assignment #2

Purpose. Lock every equation, variable name, timing convention, and solver output *before* any code is written. This sheet is the single source of truth that every teammate and every AI prompt should reference.

1. How to use this sheet

1. Open the course notes alongside this document.
2. Confirm that each equation below matches the accessible lecture PDFs and the assignment brief.
3. Any entry explicitly labeled “not numerically specified in accessible notes” should be treated as a coding calibration choice, not as a lecture fact.
4. During coding and demo prep, reference this sheet instead of re-deriving from memory.
5. If any equation changes after review, update this sheet first, then update the code.

2. Shared requirements across all models

These apply identically to Models 1, 2, and 3.

Requirement	Detail
Horizon	Infinite
Solution method	Value Function Iteration (VFI)
Shock process	Two-state Markov chain per model
Grid density	Fine grid, approximately 200–400 points
Convergence	$\max V_{n+1} - V_n < \varepsilon$, e.g. $\varepsilon = 10^{-6}$
Simulation	At least 100 periods
Moments	Mean, variance, lag-1 autocorrelation, correlation with income/output
Forecasting	Exact non-linear forecast from solved policy functions; user-selected future shocks
Dashboard	Streamlit, sidebar sliders, dark/navy styling, white chart cards, Garamond-style font
Summaries	Deterministic f-string text generated from live simulation moments

3. Model 1: Stochastic consumption-savings

3.1. Purpose

Household chooses how much to save in a risk-free asset when income randomly switches between a bad and a good state.

3.2. Bellman equation

Assignment-required infinite-horizon Bellman translation of the lecture’s two-period expected-utility problem:

$$V(a_t, y_t) = \max_{a_{t+1}} \left\{ u(c_t) + \beta \sum_{y_{t+1}} P(y_{t+1}|y_t) V(a_{t+1}, y_{t+1}) \right\}$$

Accessible lecture notes present the same economics in two-period bond notation:

$$\max_{C_t, C_{t+1}} U(C_t) + \beta E_t U(C_{t+1})$$

subject to

$$C_t + B_t = Y_t, \quad C_{t+1} = E(Y_{t+1}) + (1 + r_t)B_t.$$

Use the Bellman form above for coding, but keep the lecture notation in mind when naming variables.

3.3. Budget constraint

$$c_t + a_{t+1} = y_t + (1 + r_t) a_t, \quad c_t > 0$$

Lecture notes write saving through a non-state-contingent bond B_t with gross return $(1 + r_t)$. In the recursive implementation, the asset state a_t is the same savings vehicle written in grid form.

3.4. Utility function

CES / CRRA form stated in the assignment:

$$u(c) = \begin{cases} \frac{c^{1-\sigma} - 1}{1 - \sigma}, & \sigma \neq 1 \\ \ln(c), & \sigma = 1 \end{cases}$$

3.5. State, control, and shock identification

Field	Value
State variables	Current assets a and current income shock y
Control variable	Next-period assets a_{t+1}
Residual variable	Consumption c_t (determined from the budget constraint)
Shock	Exogenous income $y \in \{y_L, y_H\}$
Shock interpretation	Low versus high endowment

3.6. Transition matrix

$$P = \begin{pmatrix} \pi_{LL} & \pi_{LH} \\ \pi_{HL} & \pi_{HH} \end{pmatrix} \quad \text{where row } i \text{ sums to } 1.$$

Assignment-required two-state Markov discretization of the lecture's low/high income uncertainty:

$$P = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix}$$

Lecture notes confirm a two-state uncertainty environment with income outcomes Y_L and Y_H and probabilities p_L and $p_H = 1 - p_L$. They do *not* provide a recursive 2-by-2 Markov matrix, so the matrix above is the project's implementation discretization of that lecture setup and can be adjusted by sliders.

3.7. Timing convention

- Period t : agent enters with assets a_t and observes income y_t .
- Agent chooses a_{t+1} .
- Consumption is determined residually: $c_t = (1 + r_t)a_t + y_t - a_{t+1}$.

3.8. Feasibility conditions

- $c_t > 0$ always.
- $a_{t+1} \in [a_{\min}, a_{\max}]$.
- If a borrowing constraint exists: $a_{t+1} \geq a_{\min}$.

3.9. Parameters

Symbol	Meaning	Coding default / note status
β	Discount factor	0.95
σ	Risk aversion (CRRA coefficient)	2.0
r	Risk-free interest rate	0.03
y_L	Low income state	Implementation calibration chosen to represent the lecture's Y_L state
y_H	High income state	Implementation calibration chosen to represent the lecture's Y_H state
$a_{\min}$	Lower bound of asset grid	0.0
$a_{\max}$	Upper bound of asset grid	20.0
n_a	Number of asset grid points	200–400

Calibration policy for Model 1. Treat Y_L and Y_H as lecture-confirmed symbols, the infinite-horizon Bellman form as an assignment-required translation, and the numerical choices for (y_L, y_H, P) as implementation calibrations unless another lecture file provides explicit numbers.

3.10. Solver must return

- Value function $V(a, y)$: array of shape $(n_a, 2)$.
- Policy indices: array of shape $(n_a, 2)$.
- Savings policy $a_{t+1}(a_t, y_t)$: array of shape $(n_a, 2)$.
- Consumption policy $c(a, y)$: array of shape $(n_a, 2)$.
- Convergence diagnostics: iterations, final error, converged flag.
- Grid: the asset grid array.

3.11. Simulate, forecast, and plot

- Simulate: a_t, c_t, y_t for $T \geq 100$ periods.
- Forecast: user picks future shocks; apply policy to generate deterministic path.
- Plot: consumption path, asset path, income path, policy functions $a_{t+1}(a_t, y_t)$ and $c(a_t, y_t)$.
- Moments: mean, variance of c_t and a_t ; autocorrelation of c_t ; correlation of c_t with y_t .

3.12. Course-note fidelity check

Resolved from the accessible lecture notes:

- The notes use subscripts such as C_t , Y_t , and B_t , plus the gross return $(1 + r_t)$.
- The notes describe an endowment/income household that saves through a non-state-contingent bond.
- The uncertainty discussion gives two income outcomes Y_L and Y_H and expected utility $E_t U(C_{t+1})$.
- The accessible notes do not state a borrowing constraint or numerical calibration for y_L , y_H , or the transition matrix.

4. Model 2: Stochastic Robinson Crusoe economy

4.1. Purpose

Household accumulates productive capital instead of a risk-free asset. Output depends on capital and a stochastic TFP shock.

4.2. Bellman equation

Assignment-required infinite-horizon Bellman translation of the lecture's Robinson Crusoe timing:

$$V(k_t, z_t) = \max_{k_{t+1}} \left\{ u(c_t) + \beta \sum_{z_{t+1}} P(z_{t+1}|z_t) V(k_{t+1}, z_{t+1}) \right\}$$

Locked-in timing. The lecture notes explicitly say the consumer moves resources forward by choosing K_{t+1} and that tomorrow's output is defined by

$$Y_{t+1} = F(K_{t+1}).$$

The Bellman equation above is the assignment-required stochastic infinite-horizon translation of that lecture structure.

4.3. Resource constraint

$$c_t + k_{t+1} = y_t + (1 - \delta) k_t, \quad y_{t+1} = z_{t+1} F(k_{t+1}), \quad c_t > 0$$

Lecture notes write the two-period resource constraints as

$$C_t + K_{t+1} = Y_t + (1 - \delta) K_t, \quad C_{t+1} = F(K_{t+1}) + (1 - \delta) K_{t+1}.$$

With the assignment's TFP shock added, the coding convention is: current output is determined by installed capital today, while the choice of k_{t+1} determines tomorrow's output through $y_{t+1} = z_{t+1} F(k_{t+1})$.

4.4. Production function

Cobb-Douglas (stated in the assignment context):

$$F(k) = A k^\alpha$$

4.5. Utility function

Same CRRA form as Model 1:

$$u(c) = \begin{cases} \frac{c^{1-\sigma} - 1}{1-\sigma}, & \sigma \neq 1 \\ \ln(c), & \sigma = 1 \end{cases}$$

The accessible lecture notes impose increasing and concave household utility and use the same consumption theory structure as Model 1, but they do not pin down a new numeric calibration here. Keep the CRRA/log specification above as the coding parameterization unless another lecture note overrides it.

4.6. State, control, and shock identification

Field	Value
State variables	Current capital k_t and current TFP shock z_t
Control variable	Next-period capital k_{t+1}
Residual variable	Consumption c_t (from the resource constraint)
Derived variables	Output $Y_t = z_t F(k_t)$, investment $I_t = k_{t+1} - (1 - \delta)k_t$
Shock	TFP $z \in \{z_L, z_H\}$
Shock interpretation	Bad versus good productivity

Lecture timing is: current output comes from already-installed capital, while the household's choice of k_{t+1} determines next-period output. In the stochastic project version, store current output as $y_t = z_t F(k_t)$ after the state is realized, and carry forward the rule $y_{t+1} = z_{t+1} F(k_{t+1})$.

4.7. Transition matrix

$$P^z = \begin{pmatrix} \pi_{LL}^z & \pi_{LH}^z \\ \pi_{HL}^z & \pi_{HH}^z \end{pmatrix}$$

Assignment-required two-state discretization of the lecture's persistent TFP process:

$$P^z = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix}$$

The firm notes define TFP as an exogenous AR(1)-style process, $\ln A_t = (1 - \rho)A^* + \rho \ln A_{t-1} + \varepsilon_t^a$. Because the assignment requires a two-state Markov shock, (z_L, z_H, P^z) should be interpreted as a discrete approximation of that lecture process unless another note file gives a discrete version directly.

4.8. Timing convention

Lecture-consistent timing:

- Period t : the household enters with predetermined capital k_t and observes current TFP z_t .
- Current output is generated by installed capital; in the stochastic project version this is stored as $y_t = z_t F(k_t)$.
- The household chooses next-period capital k_{t+1} .
- That choice determines tomorrow's productive capital, so next-period output is $y_{t+1} = z_{t+1} F(k_{t+1})$.

4.9. Feasibility conditions

- $c_t > 0$ always.
- $k_{t+1} \in [k_{\min}, k_{\max}]$.
- $k_{t+1} \geq 0$ (no negative capital).

4.10. Parameters

Symbol	Meaning / note status
β	Discount factor 0.95
σ	Risk aversion 2.0
α	Capital share in production 0.36
δ	Depreciation rate 0.10
A	TFP scale (if used) 1.0
z_L	Low state of the discretized TFP process Implementation calibration for the low state of the discretized TFP process
z_H	High state of the discretized TFP process Implementation calibration for the high state of the discretized TFP process
$k_{\min}$	Lower bound of capital grid 0.01
$k_{\max}$	Upper bound of capital grid Implementation choice; choose as a multiple of deterministic steady-state capital so the policy does not hit the bound
n_k	Number of capital grid points 200–400

Calibration policy for Model 2. Treat the lecture AR(1) for TFP and the timing $Y_{t+1} = F(K_{t+1})$ as lecture facts. Treat the two-state objects (z_L, z_H, P^z) and the grid cap $k_{\max}$ as implementation choices used to make the assignment's VFI problem numerically tractable.

4.11. Solver must return

- Value function $V(k, z)$: array of shape $(n_k, 2)$.
- Policy indices: array of shape $(n_k, 2)$.
- Capital policy $k_{t+1}(k_t, z_t)$: array of shape $(n_k, 2)$.
- Consumption policy $c(k_t, z_t)$: array of shape $(n_k, 2)$.
- Convergence diagnostics.
- Grid: the capital grid array.

4.12. Simulate, forecast, and plot

- Simulate: k_t, c_t, Y_t, I_t for $T \geq 100$ periods.
- Forecast: user-selected future TFP shocks; apply policy to generate deterministic path.
- Plot: capital path, output path, consumption path, investment path, policy functions $k_{t+1}(k_t, z_t)$ and $c(k_t, z_t)$.
- Moments: mean and variance of Y_t and c_t ; autocorrelation of k_t or Y_t ; correlation of c_t with Y_t .

4.13. Course-note fidelity check

Resolved from the accessible lecture notes:

- The notes explicitly use $Y_{t+1} = F(K_{t+1})$, so the choice of K_{t+1} governs tomorrow's output.
- The lecture resource constraints are $C_t + K_{t+1} = Y_t + (1 - \delta)K_t$ and $C_{t+1} = F(K_{t+1}) + (1 - \delta)K_{t+1}$.
- Depreciation enters as undepreciated capital $(1 - \delta)K_t$.
- The accessible notes define a persistent TFP process, while the assignment is what forces a two-state Markov discretization for implementation.
- The accessible notes do not give numerical values for z_L, z_H , or a project-specific upper grid bound.

5. Model 3: Endogenous labor supply

5.1. Purpose

Household receives a time endowment ($T = 1$) and chooses how much labor to supply versus how much leisure to enjoy, with a stochastic wage.

5.2. Bellman equation

$$V(a_t, w_t) = \max_{a_{t+1}, L_t \in [0,1]} \left\{ u(c_t) + v(1 - L_t) + \beta \sum_{w_{t+1}} P(w_{t+1}|w_t) V(a_{t+1}, w_{t+1}) \right\}$$

Lecture-confirmed project version. The accessible lecture notes include labor supply *and* saving through a non-state-contingent bond, so the asset-inclusive version is the lecture-backed default for this project.

5.3. Budget constraint

$$c_t + a_{t+1} = (1 + r_t) a_t + w_t L_t, \quad c_t > 0$$

Accessible lecture notes write the two-period labor problem as

$$C_t + B_t = w_t L_t - T_t, \quad C_{t+1} = w_{t+1} L_{t+1} + (1 + r_t) B_t - T_{t+1}.$$

For the project sheet, use the zero-tax recursive version above unless the instructor explicitly wants taxes or profit income carried into the coding version.

5.4. Utility function

$$U(c_t, 1 - L_t) = u(c_t) + v(1 - L_t), \quad u'(\cdot), v'(\cdot) > 0, \quad u''(\cdot), v''(\cdot) < 0$$

The accessible lecture notes explicitly assume separable utility in consumption and leisure. They do *not* provide a numerical (ψ, ν) parameterization, so any parametric leisure block used in code should be documented as an implementation choice rather than as a lecture fact.

5.5. State, control, and shock identification

Field	Lecture-based project value
State variables	Current assets a_t and current wage shock w_t
Control variables	Next-period assets a_{t+1} and labor L_t
Residual	Consumption c_t (from the budget constraint)
Derived	Leisure $\ell_t = 1 - L_t$
Shock	Wage $w_t \in \{w_L, w_H\}$
Interpretation	Low versus high wage

Accessible lecture notes include saving/assets through the bond B_t , so the asset-inclusive version is the one locked in here.

5.6. Transition matrix

$$P^w = \begin{pmatrix} \pi_{LL}^w & \pi_{LH}^w \\ \pi_{HL}^w & \pi_{HH}^w \end{pmatrix}$$

Assignment-required two-state wage-shock discretization:

$$P^w = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix}$$

The accessible notes motivate wage-driven labor income but do not provide a discrete wage-shock calibration. Therefore (w_L, w_H, P^w) should be read as implementation choices used to satisfy the assignment's two-state Markov requirement.

5.7. Timing convention

- Period t : the household enters with assets a_t and observes wage w_t .
- Agent chooses labor L_t and next-period assets a_{t+1} .
- Consumption is determined residually from the budget constraint.
- Time endowment is $T = 1$, so leisure is $1 - L_t$.

Accessible lecture notes also assume an interior solution in which leisure stays strictly between zero and one.

5.8. Feasibility conditions

- $c_t > 0$.
- $0 \leq L \leq 1$ (labor bounded by time endowment).
- $1 - L > 0$ if utility requires strictly positive leisure.
- $a_{t+1} \in [a_{\min}, a_{\max}]$.

5.9. Parameters

Symbol	Meaning	Coding default / note status
β	Discount factor	0.95
σ	Consumption risk aversion	2.0
ψ	Weight on leisure in utility	Implementation calibration used only if the coding team chooses a parametric leisure block
ν	Leisure curvature parameter	Implementation calibration used only if the coding team chooses a parametric leisure block
r	Interest rate (if assets included)	0.03
w_L	Low wage state	Implementation calibration for the low state of the assignment's wage shock
w_H	High wage state	Implementation calibration for the high state of the assignment's wage shock
n_L	Number of labor grid points	50-100
n_a	Asset grid points (if applicable)	200-400

Calibration policy for Model 3. Treat the asset-inclusive budget structure and separable utility as lecture-confirmed. Treat any numerical leisure parameters and the discrete wage-shock values as implementation calibrations unless a later lecture note provides explicit formulas or numbers.

5.10. Solver must return

- Value function $V(a, w)$: shape $(n_a, 2)$.
- Savings policy $a_{t+1}(a_t, w_t)$.
- Labor policy $L(a, w)$.
- Consumption policy $c(a, w)$.
- Convergence diagnostics.

5.11. Simulate, forecast, and plot

- Simulate: L_t , c_t , w_t , and a_t if applicable, for $T \geq 100$ periods.
- Forecast: user-selected future wage shocks; apply policy.
- Plot: labor path, consumption path, wage path, assets path (if applicable), static intuition graphs (e.g. labor vs. wage, labor vs. ψ).
- Moments: mean and variance of L_t and c_t ; autocorrelation; correlation of c_t with wage income $w_t L_t$.

5.12. Course-note fidelity check

Resolved from the accessible lecture notes:

- The lecture version includes saving/assets through a bond B_t in addition to labor choice.
- Utility is separable in consumption and leisure: $U(c_t, 1 - L_t) = u(c_t) + v(1 - L_t)$.
- Leisure enters as $T - L_t$ with $T = 1$.
- The lecture's two-period constraints use wage income and can include lump-sum taxes; the project sheet uses the corresponding zero-tax recursive form.
- The accessible notes do not provide numerical values for ψ , ν , w_L , w_H , or wage-transition probabilities, so those remain documented calibrations.

6. Cross-model coding conventions

Agree on these variable names before writing any solver, simulation, or dashboard code.

Code name	Math	Meaning
beta	β	Discount factor
sigma	σ	CRRA risk aversion
r	r	Risk-free interest rate
alpha	α	Capital share (Model 2)
delta	δ	Depreciation rate (Model 2)
psi	ψ	Leisure weight (Model 3)
nu	ν	Leisure curvature (Model 3)
a_grid	$\{a_1, \dots, a_{n_a}\}$	Asset grid
k_grid	$\{k_1, \dots, k_{n_k}\}$	Capital grid
labor_grid	$\{L_1, \dots, L_{n_L}\}$	Labor grid
y_vals	$[y_L, y_H]$	Income shock values (Model 1)
z_vals	$[z_L, z_H]$	TFP shock values (Model 2)
w_vals	$[w_L, w_H]$	Wage shock values (Model 3)
P	P	2×2 transition matrix
V	V	Value function array
policy_idx	–	Index of optimal control on the grid
policy_level	–	Level value of optimal control
c_policy	$c(\cdot)$	Consumption policy array
shock_idx	–	Current shock state index (0 or 1)
T_sim	T	Simulation horizon
T_fcast	–	Forecast horizon
seed	–	Random seed for reproducibility

Solver return format (all models):

Every solver should return a dictionary with these keys:

- value_function
- policy_indices
- policy_levels (savings/capital/labor as appropriate)
- c_policy
- grid (the state grid used)
- diagnostics (dict with iterations, final_error, converged)

7. Before coding starts: final checklist

1. All note-dependent placeholders in this sheet have been resolved or explicitly labeled as not numerically specified in accessible notes.
2. For Model 2, the timing convention centered on $Y_{t+1} = F(K_{t+1})$ is agreed by the group.
3. For Model 3, the asset-inclusive lecture version is the one the group intends to code unless another instructor note overrides it.
4. For Model 3, the utility form is confirmed as separable unless a later course note replaces it.
5. The coding names in the convention table are agreed upon.
6. The solver return format is agreed upon.
7. Every teammate can answer: “What is the state? What is the control? What is the constraint? What is the timing?” for each model.
8. This sheet is saved where every teammate can access it during coding and demo prep.

Rule. If the course notes disagree with anything on this sheet, the course notes win. Update this sheet, then update the code.